

KodetoCareer — Product Requirements Document (PRD)

Version 1.0 | For Engineering and Design Teams

1. Document Purpose

This PRD defines what the KodetoCareer Training & Placement Management Platform must do. It is the authoritative source of truth for product decisions during MVP development. Engineering teams should treat the acceptance criteria in each section as contractual requirements. Design teams should treat the user stories as the basis for every screen.

2. Product Overview

Product Name: KodetoCareer Platform

Platform Type: Multi-tenant SaaS (Web + Mobile)

Primary Users: Students, Trainers, College Admins, KodetoCareer Super Admin

MVP Timeline: 10 weeks

MVP Goal: Onboard 3 pilot colleges with full learning and attendance management

3. User Roles & Permissions Matrix

Feature Area | Super Admin | College Admin | Trainer | Student

Create college | ■ | ■ | ■ | ■

Create batch | ■ | ■ | ■ | ■

Enroll students | ■ | View only | ■ | ■

Upload content | ■ | ■ | ■ | ■

Mark attendance | ■ | ■ | ■ | ■

Take quizzes | ■ | ■ | ■ | ■

View all colleges | ■ | ■ | ■ | ■

View own college | ■ | ■ | ■ | ■

Generate certificates | ■ | ■ | ■ | ■

Upload offer letters | ■ | ■ | ■ | ■

View placement stats | ■ | ■ | ■ | Own only

Download reports | ■ | ■ | ■ | ■

4. Module-by-Module Requirements

MODULE 1: Authentication & Onboarding

1.1 Student Registration & Login

User Story:

As a student, I want to log in with my email and password so that I can access my personalised learning dashboard.

Functional Requirements:

FR-1.1.1 — The system shall support email/password login for all user types.

FR-1.1.2 — Students shall receive a unique 8-character student ID on registration (format: KTC-YYYY-XXXXX).

FR-1.1.3 — On first login, students shall be shown a mandatory profile completion screen collecting: full name, phone number, profile photo, date of birth, current college, course enrolled.

FR-1.1.4 — The system shall support OTP-based email verification before account activation.

FR-1.1.5 — Forgotten password flow must complete within 3 steps: email entry → OTP → new password.

FR-1.1.6 — All sessions shall expire after 30 days of inactivity with a warning shown at 25 days.

FR-1.1.7 — The system shall support parallel login on one mobile device and one web browser simultaneously.

Acceptance Criteria:

- [] A new student can register, verify email, and reach their dashboard in under 2 minutes
- [] An existing student can log in within 10 seconds on a 4G connection
- [] OTP arrives within 60 seconds of request

- [] Invalid login attempts are blocked after 5 failures for 15 minutes

1.2 Admin & Trainer Onboarding

User Story:

As a Super Admin, I want to create College Admin and Trainer accounts so that I can delegate management without sharing my credentials.

Functional Requirements:

FR-1.2.1 — Super Admin shall create College Admin accounts by entering name, email, college assignment, and phone number. The system auto-generates and emails a temporary password.

FR-1.2.2 — Super Admin shall create Trainer accounts with name, email, phone, specialisation tags, and assigned colleges.

FR-1.2.3 — First login for any admin type must trigger a forced password change.

FR-1.2.4 — Admin accounts shall be deactivatable (not deletable) to preserve audit history.

MODULE 2: Course & Content Management

2.1 Course Structure

User Story:

As a Super Admin or Trainer, I want to create structured courses with modules, lessons, and resources so that students follow a logical learning path.

Functional Requirements:

FR-2.1.1 — Courses shall have a three-level hierarchy: Course → Module → Lesson.

FR-2.1.2 — Each Lesson may contain any combination of: video (MP4, YouTube link, or Vimeo link), PDF notes, external links, text content (rich text), and assignments.

FR-2.1.3 — Lessons shall be orderable via drag-and-drop in the admin panel.

FR-2.1.4 — Modules shall be lockable (student cannot access until previous module is completed).

FR-2.1.5 — A course shall have: title, description, category tag, difficulty level (Beginner/Intermediate/Advanced), estimated duration, thumbnail image, and assigned trainer(s).

FR-2.1.6 — Course publishing shall have three states: Draft, Review, Published.

FR-2.1.7 — A course may be assigned to one or more batches.

Acceptance Criteria:

- [] Admin can create a complete 5-module course with lessons in under 30 minutes
- [] Students cannot access locked modules until prerequisites are complete
- [] A 500MB video upload completes within 5 minutes on a 10 Mbps connection

2.2 Video Management**Functional Requirements:**

FR-2.2.1 — Uploaded videos shall be stored on AWS S3 and served via CloudFront CDN.

FR-2.2.2 — The system shall support two video source types: direct upload (MP4 up to 2GB) and embedded URL (YouTube, Vimeo, Google Drive).

FR-2.2.3 — Direct-upload videos shall be transcoded to adaptive bitrate (480p, 720p, 1080p) using AWS MediaConvert.

FR-2.2.4 — Video player must support: playback speed control (0.5x to 2x), quality selection, fullscreen, and picture-in-picture.

FR-2.2.5 — Student video watch progress shall be persisted so playback resumes from last position.

FR-2.2.6 — Videos shall not be downloadable by students (stream-only delivery).

FR-2.2.7 — Admin panel shall show per-video analytics: total views, average watch completion percentage.

2.3 Notes & Study Materials**Functional Requirements:**

FR-2.3.1 — Trainers and Super Admins may upload PDF notes (max 50MB each) attached to any lesson.

FR-2.3.2 — Students may download attached PDF notes.

FR-2.3.3 — The system shall maintain a "My Notes" section where students can write personal text notes against any lesson.

FR-2.3.4 — Student personal notes are private and not visible to trainers or admins.

MODULE 3: Batch & Attendance Management**3.1 Batch Management**

User Story:

As a Super Admin, I want to create batches that link students, a college, a course, and a trainer so that I can manage multiple training engagements simultaneously.

Functional Requirements:

FR-3.1.1 — A Batch shall have: batch name, start date, end date, assigned college, assigned course, assigned trainer(s), schedule (days and time), and mode (online/offline/hybrid).

FR-3.1.2 — Students shall be bulk-imported to a batch via CSV upload. Required CSV columns: Full Name, Email, Phone, Enrollment Number.

FR-3.1.3 — System shall validate CSV and show errors per row before import (duplicate email, missing fields).

FR-3.1.4 — A student may be enrolled in multiple batches simultaneously.

FR-3.1.5 — Batch capacity shall have a configurable maximum (default: 200 students).

3.2 Attendance System

User Story:

As a Trainer, I want to mark attendance for each class session so that the admin and college can see which students attended.

Functional Requirements:

FR-3.2.1 — Trainer can create a Class Session for a batch with: date, start time, end time, topic covered, and mode (online/offline).

FR-3.2.2 — Attendance marking interface shall show all enrolled students in a list. Trainer marks Present/Absent/Late for each student.

FR-3.2.3 — Trainer may mark attendance up to 24 hours after a session with a reason note.

FR-3.2.4 — Super Admin may override any attendance record with an audit log entry.

FR-3.2.5 — System shall auto-calculate attendance percentage per student per batch.

FR-3.2.6 — Students with attendance below a configurable threshold (default: 75%) shall be flagged with a visual indicator on the admin dashboard.

FR-3.2.7 — Students shall receive a push notification when attendance is marked for their batch.

FR-3.2.8 — Attendance reports shall be exportable as Excel/PDF, filterable by batch, date range, and student.

Acceptance Criteria:

- [] A trainer can mark attendance for a 100-student batch in under 5 minutes
- [] Attendance percentage is visible on the college dashboard within 1 minute of the trainer submitting

MODULE 4: Assessments (Quizzes & Assignments)

4.1 Quiz/Test Engine

User Story:

As a Trainer, I want to create timed quizzes with multiple question types so that I can assess student understanding of each module.

Functional Requirements:

FR-4.1.1 — Quiz shall support question types: Multiple Choice (single answer), Multiple Choice (multiple answer), True/False, and Short Text Answer.

FR-4.1.2 — Quiz settings: total marks, pass marks, time limit (optional), number of attempts allowed, shuffle questions, shuffle options.

FR-4.1.3 — Questions shall be tagged with: topic tag, difficulty (Easy/Medium/Hard), and marks value.

FR-4.1.4 — A question bank shall store all questions by course/topic. Trainers can create new quizzes by picking from the question bank.

FR-4.1.5 — During an active quiz, the student's progress (current question, time remaining) shall auto-save every 30 seconds to prevent data loss on network interruption.

FR-4.1.6 — On quiz submission, the system shall immediately display the score and correct answers (configurable: show answers immediately or after deadline).

FR-4.1.7 — Leaderboard for each quiz showing top 10 performers shall be visible to all batch students after the deadline.

FR-4.1.8 — Admin and Trainer can view per-question analytics: what percentage of students answered correctly.

4.2 Assignments

Functional Requirements:

FR-4.2.1 — Assignments shall have: title, description, deadline, total marks, and file submission option.

FR-4.2.2 — Students submit assignments as file upload (PDF, ZIP, DOCX) with an optional text note.

FR-4.2.3 — Trainer reviews submissions and enters marks with written feedback.

FR-4.2.4 — Students are notified when their assignment is graded.

FR-4.2.5 — Late submissions are accepted but marked as "Late" with a timestamp.

MODULE 5: Progress Tracking & Analytics

5.1 Student Progress Dashboard

Functional Requirements:

FR-5.1.1 — Student dashboard shall show: overall course completion percentage, attendance rate, quiz average score, assignment completion rate, current streak (consecutive days of activity), and placement readiness score (composite metric).

FR-5.1.2 — A per-module progress breakdown shall show completion status of every lesson and assessment.

FR-5.1.3 — A "Weekly Activity" heatmap (GitHub-style) shall show study activity over the last 12 weeks.

FR-5.1.4 — Students shall see a "Skill Level" indicator per topic (Beginner/Intermediate/Advanced) computed from quiz scores.

5.2 Admin Analytics Dashboard

Functional Requirements:

FR-5.2.1 — Super Admin dashboard shall show: total active students, total active colleges, total active batches, students placed this month, average attendance across all batches, and revenue summary.

FR-5.2.2 — College-level drill-down shows all metrics for a single college.

FR-5.2.3 — Admin can filter any report by: college, batch, date range, course, and trainer.

FR-5.2.4 — A "Students at Risk" report flags students with: attendance < 75%, quiz average < 40%, or zero activity in the last 14 days.

FR-5.2.5 — All reports shall be exportable to PDF and Excel.

MODULE 6: Certificates & Documents

6.1 Certificate Generation

User Story:

As a Super Admin, I want to generate completion certificates for students who finish a course and meet the minimum criteria so that they have verifiable proof of their training.

Functional Requirements:

FR-6.1.1 — Certificate eligibility criteria (configurable per course): minimum attendance percentage, minimum quiz average score, all mandatory assignments submitted.

FR-6.1.2 — Certificates shall be generated as PDF with: student name, course name, completion date, unique certificate ID (QR code), trainer name, and KodetoCareer authorised signature and seal.

FR-6.1.3 — Certificate ID shall be verifiable at a public URL (no login required):

`verify.kodetocareer.com/[certificate-id]`

FR-6.1.4 — Student receives the certificate in their Student Portal under "My Certificates" and via email.

FR-6.1.5 — Students can share their certificate as a direct link to LinkedIn or via WhatsApp.

FR-6.1.6 — Bulk certificate generation for an entire batch shall complete within 5 minutes.

6.2 Offer Letter & Internship Letter Storage

Functional Requirements:

FR-6.2.1 — Super Admin may upload offer letters and internship letters against a specific student's profile.

FR-6.2.2 — Student sees their uploaded letters under "My Documents" in the portal.

FR-6.2.3 — Letters are stored as PDFs. Students can download but not delete.

FR-6.2.4 — Placement record associated with an offer letter requires: company name, role/designation, package (annual CTC), offer date, joining date, and offer type (Internship/Full-Time/Contract).

MODULE 7: Placement Management

7.1 Placement Tracking

Functional Requirements:

FR-7.1.1 — Each student has a placement status: Not Started | Preparing | Actively Applying | Placed | On Hold.

FR-7.1.2 — Admin can log placement activity: interview rounds, offer received, offer accepted.

FR-7.1.3 — Placement dashboard shows: total students placed this year, average package, company-wise breakdown, and month-wise trend chart.

FR-7.1.4 — College Admin can see placement statistics for students from their college only.

FR-7.1.5 — A "Placement Leaderboard" shows top placed students (visible to all students in their college with the student's consent).

7.2 Job Opportunities Board

Functional Requirements:

FR-7.2.1 — Admin can post job/internship opportunities with: company name, role, skills required, CTC/stipend, location, application deadline, and application link or email.

FR-7.2.2 — Students see the job board filtered to opportunities matching their skill level.

FR-7.2.3 — Students can mark interest ("I'm interested") on a job post, which notifies the admin.

FR-7.2.4 — Job posts expire automatically after the deadline date.

MODULE 8: Communication System

Functional Requirements:

FR-8.1 — Super Admin and Trainers can send announcements to: all students, all students in a specific batch, all students in a specific college, or an individual student.

FR-8.2 — Announcements appear in the app Notification Centre and trigger push notifications.

FR-8.3 — System shall send automated email notifications for: new lesson available, quiz deadline approaching (24 hours before), assignment graded, certificate ready, attendance below threshold (to student and college admin).

FR-8.4 — A bulk SMS option (via Twilio or MSG91) shall be available for critical communications.

FR-8.5 — WhatsApp notification integration (via WhatsApp Business API or Interakt) shall be included in Version 2.

MODULE 9: College Dashboard (Partner Portal)

Functional Requirements:

FR-9.1 — College Admin login shows a dashboard scoped to their college only.

FR-9.2 — College dashboard shows: enrolled student count, active batch count, overall attendance rate, average quiz score, students placed, and pending actions (flagged students).

FR-9.3 — College Admin can view individual student profiles (read-only).

FR-9.4 — College Admin can download attendance reports, progress reports, and placement reports as PDF/Excel.

FR-9.5 — College Admin cannot modify any student data — read and download only.

FR-9.6 — College Admin receives a weekly automated email summary of their college's statistics.

5. Non-Functional Requirements

Requirement | Target

Page load time (web) | < 3 seconds on 4G

API response time | < 500ms for 95th percentile

Mobile app launch time | < 2 seconds (cold start)

System uptime | 99.5% monthly

Concurrent users supported | 10,000 in MVP, scalable to 100,000

Data backup frequency | Daily automated backup to separate AWS region

Video buffering | < 5 seconds to start playback on 4G

File upload progress | Real-time progress indicator for uploads > 1MB

Maximum supported file upload | 2GB for video, 50MB for PDF

6. MVP Scope vs. Future Scope

Feature | MVP (v1) | Version 2 | Version 3

Student login & dashboard | ■

Course & content management | ■

Attendance management | ■

Quizzes & assignments | ■

Certificate generation | ■

Admin CRM panel | ■

College dashboard | ■

Placement tracking (basic) | ■

Job board | ■

Push notifications | ■

Resume builder | ■

AI mock interview | ■

Coding assessment platform | ■

WhatsApp automation | ■

Recruiter portal | ■

AI career guidance | ■

AI coding mentor | ■

Multi-language support | ■

Third-party training partner licensing | ■

7. Success Metrics for MVP Launch

Metric | Target at 3 months post-launch

Colleges onboarded | 3 pilot colleges

Active students | 500+

Daily Active Users (students) | 40%+ of enrolled

Quiz completion rate | > 70%

Attendance marking compliance | 100% of sessions recorded

Certificates generated | > 90% of eligible students

Admin satisfaction score | > 4.0 / 5.0

Student satisfaction score | > 4.2 / 5.0

End of Product Requirements Document

Next document: Technical Requirements Document (TRD)